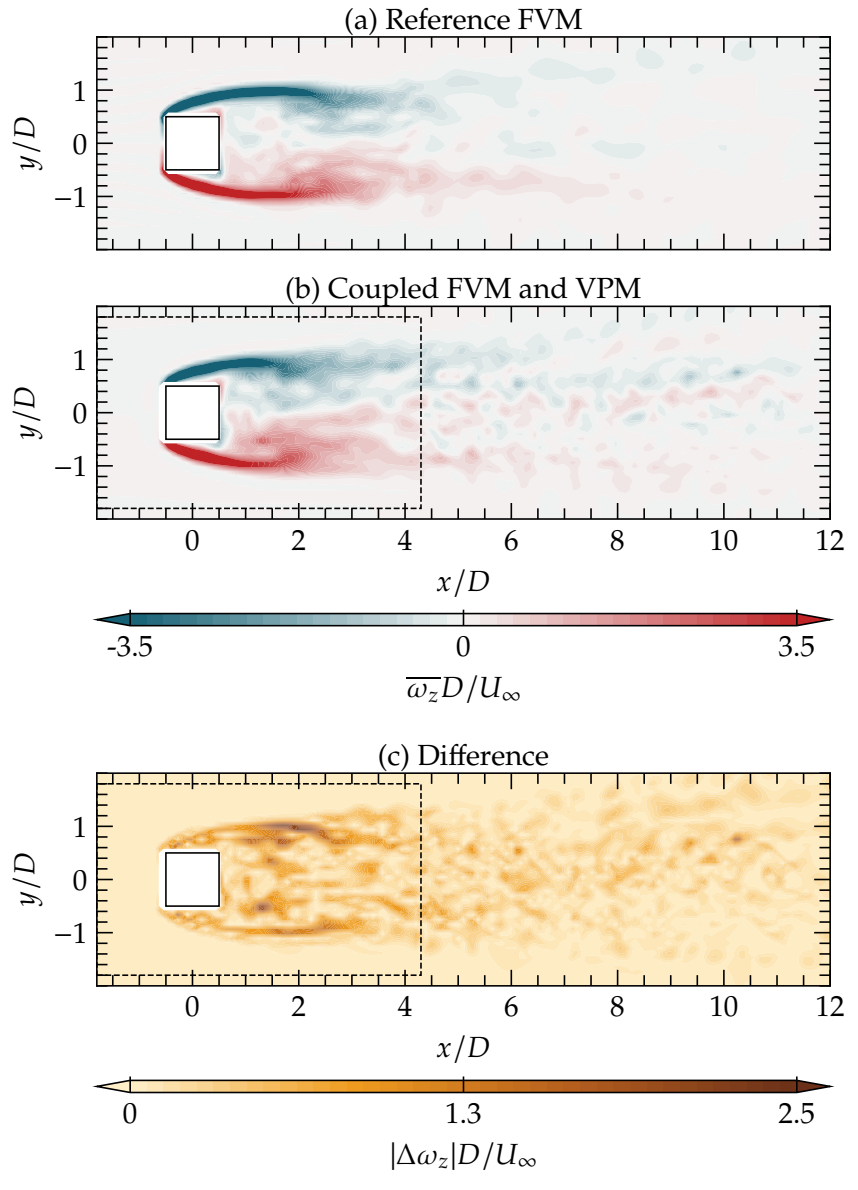

(a) Reference FVM
(b) Coupled FVM and VPM
(c) Difference
y/D
x/D
1
0
−1
0
2
4
6
8
10
12
-3.5
0
3.5
$\overline{\omega_z} D/U_\infty$
0
1.3
2.5
$|\Delta\omega_z| D/U_\infty$